

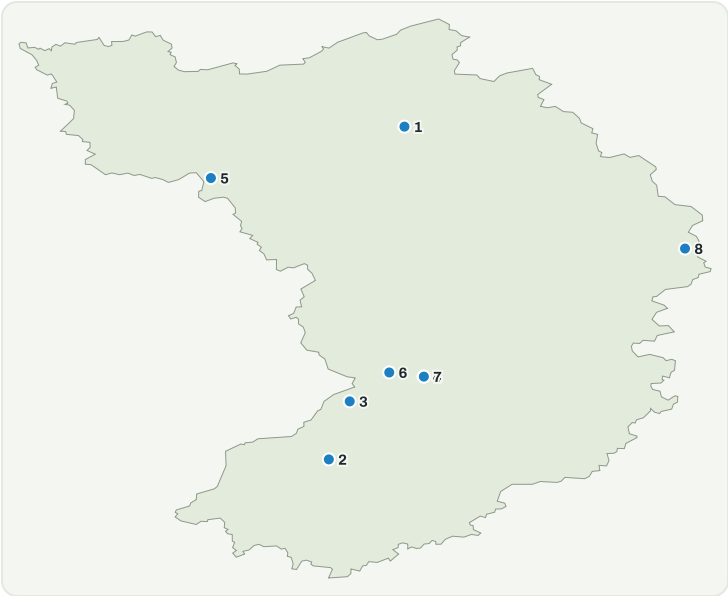
Which culverts to replace: Upper French Broad watershed

Budget \$2.00M. Priority: rivers only. Generated October 3, 2026 from the National Aquatic Barrier Inventory and USGS NHDPlus HR.

8 projects (8 culverts)	\$2.00M total planning cost	46.6 mi of creek reconnected	15% of washout risk removed	4.0 mi of brook trout habitat reopened
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With the same money, picking culverts one at a time would reconnect 39.6 mi and remove 17% of washout risk. This plan reconnects **7.0 more miles** (+18%).

Under 30 random cost scenarios (each culvert's cost varied by up to about 2× either way), this approach beat one-at-a-time ranking in **30**. Each project's confidence is how often it stayed in the plan.



Projects are numbered in priority order.

How projects are ordered
By benefit per dollar: river miles (and washout risk, when weighted) per \$1M. Do them top to bottom if money arrives in stages. Culverts in one project must be done together.

Projects, in priority order

1 <i>Hunter Creek, Madison County</i> Replace 1 culvert			
Cost \$274k likely \$121k–\$625k	River reconnected 8.1 mi	Washout risk removed 1.8% of the watershed total	Confidence Medium kept in 63% of scenarios
What to build 1. Hunter Creek: replace the pipe under an unnamed road with a creek-wide crossing about 26 ft wide (a small bridge or bottomless arch, natural creek bed). \$274k. 35.87293, -82.67872, inventory ID sm54722 Why it's in the plan Strong on its own: 8.1 mi of creek reconnected for \$274k. Next steps ☐ Site visit: confirm the pipe size and condition (the inventory has no width measurement for this culvert). ☐ Engineering estimate: the cost here is a planning figure; real costs often differ by up to 2×. ☐ Identify the road owner (not recorded in the inventory).			
2 <i>Tributary of Davidson River, Transylvania County</i> Replace 1 culvert			
Cost \$169k likely \$74k–\$385k	River reconnected 4.7 mi	Washout risk removed 1.1% of the watershed total	Confidence Medium kept in 77% of scenarios
What to build 1. Tributary of Davidson River: replace the pipe under 475C (unpaved) with a creek-wide crossing about 22 ft wide (a small bridge or bottomless arch, natural creek bed). \$169k. 35.27755, -82.78085, inventory ID sm164711, road owner: USDA Forest Service Why it's in the plan Good value: 4.7 mi of creek for \$169k, about 28 mi per \$1M. Ranking culverts one at a time doesn't pick it at this budget. Next steps			

- ☐ Site visit: confirm the pipe size and condition.
- ☐ Engineering estimate: the cost here is a planning figure; real costs often differ by up to 2×.
- ☐ Coordinate with the road owner: USDA Forest Service.

3 *Unnamed tributary, Transylvania County*

Replace 1 culvert

Cost \$164k likely \$72k–\$374k	River reconnected 4.3 mi 2.0 mi brook trout habitat	Washout risk removed 1.7% of the watershed total	Confidence High kept in 80% of scenarios
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What to build

1. **Unnamed tributary:** replace the pipe under W1206 (unpaved) with a creek-wide crossing about 21 ft wide (a small bridge or bottomless arch, natural creek bed). \$164k.

[35.38150, -82.75263](#), inventory ID [sm59186](#), road owner: USDA Forest Service

Why it's in the plan

Good value: 4.3 mi of creek for \$164k, about 26 mi per \$1M. Ranking culverts one at a time doesn't pick it at this budget.

Next steps

- ☐ Site visit: confirm the pipe size and condition.
- ☐ Engineering estimate: the cost here is a planning figure; real costs often differ by up to 2×.
- ☐ Coordinate with the road owner: USDA Forest Service.
- ☐ Native brook trout habitat upstream (2.0 mi): bring in NC Wildlife Resources Commission biologists.

4 *Wash Creek, Henderson County*

Replace 1 culvert

Cost \$481k likely \$212k–\$1.10M	River reconnected 11.4 mi	Washout risk removed 5.1% of the watershed total	Confidence Medium kept in 73% of scenarios
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What to build

1. **Wash Creek:** replace the pipe under 142 (paved) with a creek-wide crossing about 37 ft wide (a small bridge or bottomless arch, natural creek bed). \$481k.

[35.42579, -82.65310](#), inventory ID [sm3043](#), road owner: USDA Forest Service

Why it's in the plan

Strong on its own: 11.4 mi of creek reconnected for \$481k.

High washout risk (score 0.47 of 1): moderate pipe constriction on a paved road draining 2.8 sq mi.

Next steps

- ☐ Site visit: confirm the pipe size and condition.
- ☐ Engineering estimate: the cost here is a planning figure; real costs often differ by up to 2×.
- ☐ Coordinate with the road owner: USDA Forest Service.

5

East Fork Little Creek, Madison County

Replace 1 culvert

Cost

\$234k

likely \$103k–\$534k

River reconnected

4.9 mi

1.5 mi brook trout habitat

Washout risk removed

1.3%

of the watershed total

Confidence

Medium

kept in 60% of scenarios

What to build

1. **East Fork Little Creek:** replace the pipe under an unnamed road with a creek-wide crossing about 22 ft wide (a small bridge or bottomless arch, natural creek bed). \$234k.

[35.78113](#), [-82.94035](#), inventory ID [sm2625](#), road owner: Local government

Why it's in the plan

Good value: 4.9 mi of creek for \$234k, about 21 mi per \$1M. Ranking culverts one at a time doesn't pick it at this budget.

Next steps

- ☐ Site visit: confirm the pipe size and condition (the inventory has no width measurement for this culvert).
- ☐ Engineering estimate: the cost here is a planning figure; real costs often differ by up to 2×.
- ☐ Coordinate with the road owner: Local government.
- ☐ Native brook trout habitat upstream (1.5 mi): bring in NC Wildlife Resources Commission biologists.

6

Middle Fork, Henderson County

Replace 1 culvert

Cost

\$199k

likely \$88k–\$454k

River reconnected

4.0 mi

0.5 mi brook trout habitat

Washout risk removed

1.3%

of the watershed total

Confidence

Medium

kept in 57% of scenarios

What to build

1. **Middle Fork:** replace the pipe under Fs5097 (unpaved) with a creek-wide crossing about 27 ft wide (a small bridge or bottomless arch, natural creek bed). \$199k.

[35.43299](#), [-82.69930](#), inventory ID [sm14723](#), road owner: USDA Forest Service

Why it's in the plan

Good value: 4.0 mi of creek for \$199k, about 20 mi per \$1M. Ranking culverts one at a time doesn't pick it at this budget.

Next steps

- ☐ Site visit: confirm the pipe size and condition (the inventory has no width measurement for this culvert).
- ☐ Engineering estimate: the cost here is a planning figure; real costs often differ by up to 2×.
- ☐ Coordinate with the road owner: USDA Forest Service.
- ☐ Native brook trout habitat upstream (0.5 mi): bring in NC Wildlife Resources Commission biologists.

7 *Bear Branch, Henderson County*

Replace 1 culvert

Cost
\$245k

likely \$108k–\$559k

River reconnected
4.9 mi

Washout risk removed
1.4%
of the watershed total

Confidence
Medium
kept in 53% of scenarios

What to build

1. **Bear Branch:** replace the pipe under 5000 with a creek-wide crossing about 23 ft wide (a small bridge or bottomless arch, natural creek bed). \$245k.

[35.42558](#), [-82.65247](#), inventory ID [sm168304](#), road owner: USDA Forest Service

Why it's in the plan

Good value: 4.9 mi of creek for \$245k, about 20 mi per \$1M. Ranking culverts one at a time doesn't pick it at this budget.

Next steps

- ☐ Site visit: confirm the pipe size and condition (the inventory has no width measurement for this culvert).
- ☐ Engineering estimate: the cost here is a planning figure; real costs often differ by up to 2×.
- ☐ Coordinate with the road owner: USDA Forest Service.

8 *Big Piney Branch, Buncombe County*

Replace 1 culvert

Cost
\$214k

likely \$94k–\$488k

River reconnected
4.2 mi

Washout risk removed
1%
of the watershed total

Confidence
Medium
kept in 53% of scenarios

What to build

1. **Big Piney Branch:** replace the pipe under an unnamed road with a creek-wide crossing about 19 ft wide (a small bridge or bottomless arch, natural creek bed). \$214k.

35.65493, -82.29942, inventory ID [sm54721](#)

Why it's in the plan

Good value: 4.2 mi of creek for \$214k, about 20 mi per \$1M. Ranking culverts one at a time doesn't pick it at this budget.

Next steps

- ☐ Site visit: confirm the pipe size and condition (the inventory has no width measurement for this culvert).
- ☐ Engineering estimate: the cost here is a planning figure; real costs often differ by up to 2×.
- ☐ Identify the road owner (not recorded in the inventory).

How this plan was made

Pinchpoint treats each river as a network. A culvert only reconnects the creek above it once every blocking culvert below it is also fixed, and the gain is capped by how much open river lies below (the inventory's own rule). An exact optimizer picks the set of culverts with the most benefit for the budget, weighing reconnected river miles against a washout-risk score (pipe constriction × stream size × road type) as you set it.

Limits: costs are planning estimates from stream width and road type; the washout score is a screening index, not a failure prediction; only culverts that field crews have assessed are included; dams and waterfalls are treated as permanent.